

Dimensions of permanental varieties in arbitrary size

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Abstract

Let $I_h(a, b)$ be the ideal generated by the $h \times h$ permanents of a generic $a \times b$ matrix over a field of characteristic different from two. We prove that $\dim K[X]/I_h(a, b) = \max(a, b)(h - 1)$ whenever $1 \leq h \leq \min(a, b)$ and $h < \max(a, b)$. Consequently, the critical locus of the permanent of order n has codimension $2n$ for every $n \geq 2$. We also prove that the maximal permanents of a generic $k \times (k + 1)$ matrix generate a geometrically reduced complete intersection. The dimension argument uses a pivot expansion and cubic relations in the border variables. Two successive comparisons with highest-degree equations reduce the relevant fibers to a coordinate-support calculation. Reducedness follows from a separate regular-sequence argument and a dimension estimate excluding components over the exceptional locus.

1 Introduction

For a matrix $X = (x_{ij})$ of independent variables, the permanent of a square submatrix is the sum of its transversal monomials, all with coefficient one. Write $I_h(a, b) \subset K[x_{ij}]$ for the ideal of all $h \times h$ permanents of a generic $a \times b$ matrix, and write $V_h(a, b)$ for its zero locus over an algebraic closure. All dimensions are affine dimensions. In particular, codimension is measured in the full matrix space, rather than in a hypersurface.

Boralevi, Carlini, Michałek, and Ventura studied the codimensions of maximal permanental varieties and of the singular locus of the permanental hypersurface [1]. Their Conjecture 3.4 asserts that the $k + 1$ maximal permanents of a generic $k \times (k + 1)$ matrix define a complete intersection in characteristic zero. Their Theorem 3.18 establishes the expected codimension for maximal permanents with $k \leq 5$. Earlier work of Kirkup treats minimal primes and the 3×4 case in detail [2, Section 7].

The following result gives the dimension for every proper subpermanent size, including maximal permanents of nonsquare matrices.

Theorem 1.1. *Let K be a field with $\text{char } K \neq 2$. If $1 \leq h \leq \min(a, b)$ and $h < \max(a, b)$, then*

$$\dim K[X]/I_h(a, b) = \max(a, b)(h - 1). \quad (1.1)$$

Equivalently,

$$\text{ht } I_h(a, b) = \max(a, b)(\min(a, b) - h + 1).$$

The condition $h < \max(a, b)$ excludes the full permanent of a square matrix, which defines a hypersurface. Taking $a = b = n$ and $h = n - 1$ gives the exact critical-locus dimension.

Corollary 1.2. *For $n \geq 2$ and $\text{char } K \neq 2$,*

$$\text{codim}_{\mathbb{A}^{n^2}} \text{Crit}(\text{per}_n) = 2n.$$

The same equality holds for the singular locus of $\{\text{per}_n = 0\}$, with codimension still measured in $\mathbb{A}^{n^2}$.

The dimension calculation alone does not imply reducedness of the ideals. For maximal permanents of a matrix with one more column than rows, a further argument gives the following scheme-theoretic result.

Theorem 1.3. *For every $k \geq 1$ and every field K of characteristic different from two, $I_k(k, k+1)$ is a geometrically radical complete-intersection ideal of height $k+1$.*

Here geometrically radical means that the ideal remains radical after every field extension. In particular, Theorem 1.3 proves the complete-intersection assertion of [1, Conjecture 3.4] and also gives reducedness in every characteristic other than two.

The prime-size preprint [4] proves a Frobenius-splitting statement by an explicit trace calculation in the relevant prime characteristic. The present paper gives a separate dimension proof valid at every size; no recursion from a prime size is used. Its conclusions about dimension and reducedness do not supply a Frobenius splitting. The arithmetic distinction is already visible for the 3×4 ideal, whose Frobenius behavior is studied in [5].

The proof of Theorem 1.1 is in Sections 2–3. After normalizing a nonzero matrix entry, one first takes highest-degree parts in the interior block. Cubic relations in the border survive this step. A second highest-degree comparison, made after fixing a smaller nonzero subpermanent, bounds the border fibers. Section 5 proves reducedness, with the required filtered-algebra lemma supplied in Section 4. Section 6 records consequences and the characteristic-two boundary.

2 A pivot identity and two elementary lemmas

We work over an algebraically closed field until otherwise stated. For row and column sets A, B of the same size, set

$$p_{A,B} = \text{per } Y[A, B], \quad p_{\emptyset, \emptyset} = 1.$$

Whenever a pivot entry is nonzero, multiplying its row by the inverse of that entry identifies the chart with $\mathbb{G}_m$ times a chart whose pivot is one. Row and column permutations allow us to place the pivot in the lower-right corner. The normalized matrix has the form

$$X = \begin{pmatrix} Y & u \\ v^\top & 1 \end{pmatrix}. \quad (2.1)$$

These operations preserve the vanishing of all permanents of a fixed size. For $|A| = |B|$, denote by $P_{A,B}$ the permanent obtained by adjoining the pivot row and column to $Y[A, B]$. Direct expansion gives

$$P_{A,B} = p_{A,B} + \sum_{i \in A, j \in B} u_i v_j p_{A \setminus \{i\}, B \setminus \{j\}}. \quad (2.2)$$

If $|A| = h-1$ and $|B| = h$, let $R_{A,B}$ be the $h \times h$ permanent using rows A and the pivot row, and columns B , without the pivot column.

Lemma 2.1 (Cubic border relation). *For these sets A, B ,*

$$\sum_{j \in B} v_j P_{A, B \setminus \{j\}} - R_{A, B} = 2 \sum_{\substack{i \in A \\ \{j, \ell\} \subset B}} u_i v_j v_\ell P_{A \setminus \{i\}, B \setminus \{j, \ell\}}. \quad (2.3)$$

The sum on the right runs over unordered two-element column sets. There is a transposed identity with two u variables and one v variable.

Proof. Expand every P using (2.2). Its first terms sum to $\sum_{j \in B} v_j P_{A, B \setminus \{j\}} = R_{A, B}$. In the remaining sum, the two distinct columns j, ℓ occur in both orders. Their two contributions are equal, giving the factor two. Transposing the matrix gives the second identity. $\square$

Lemma 2.2 (Highest-degree comparison). *Let $f_1, \dots, f_s \in K[y_1, \dots, y_d, z_1, \dots, z_e]$. Let q_i be the nonzero part of f_i of highest total degree in the y variables, with zero polynomials omitted. Then*

$$\dim V(f_1, \dots, f_s) \leq \dim V(q_1, \dots, q_s).$$

The variables z have weight zero and need not be specialized.

Proof. Let Z be an irreducible component of $V(f_1, \dots, f_s)$ of dimension d_Z . In $\mathbb{A}^{d+e} \times \mathbb{A}^1$, take the closure $\mathcal{Z}$ of

$$\{(ty, z, t) : (y, z) \in Z, t \neq 0\}.$$

It is irreducible of dimension $d_Z + 1$. Its fiber at $t = 0$ is nonempty: scaling any fixed point of Z gives a curve with a limit at $t = 0$. The coordinate t is neither zero nor a unit on $\mathcal{Z}$. The principal ideal theorem and the dimension formula for affine domains over a field therefore give $\dim(\mathcal{Z} \cap \{t = 0\}) \geq d_Z$. If $d_i = \deg_y f_i$, the polynomial $t^{d_i} f_i(y/t, z)$ vanishes on $\mathcal{Z}$ and restricts to q_i at $t = 0$. Thus that fiber lies in $V(q_1, \dots, q_s)$. Taking the maximum over Z proves the assertion. This is the usual weighted homogenization proof of the filtered dimension comparison. $\square$

Lemma 2.3 (The border support model). *For $k \geq 2$, the zero locus in $\mathbb{A}^{2k}$ of*

$$u_i v_j v_\ell = 0 \quad (j \neq \ell), \quad u_i u_{i'} v_j = 0 \quad (i \neq i') \quad (2.4)$$

has dimension k .

Proof. If either $u = 0$ or $v = 0$, the other vector is arbitrary, giving two k -dimensional coordinate spaces. If both vectors are nonzero, the first family forces v to have at most one nonzero coordinate, and the second forces the same for u . This part is contained in the union of the coordinate planes on one u coordinate and one v coordinate. The whole locus is therefore a finite union of spaces of dimensions k and two, with maximum k . $\square$

3 The dimension theorem

Proof of Theorem 1.1 for square matrices. We prove simultaneously for all $1 \leq h < n$ that

$$\dim V_h(n, n) = n(h - 1). \quad (3.1)$$

The case $h = 1$ is immediate, and starts the induction on n . Suppose $2 \leq h < n$, set $m = n - 1$, and use the normalized pivot chart (2.1), with Y of size $m \times m$. Put

$$r = h - 2, \quad k = m - r = n - h + 1 \geq 2.$$

The normalized ambient space has dimension $m^2 + 2m$.

Give entries of Y weight one and all border variables weight zero. For $|A| = |B| = h - 1$, the highest Y -degree part of $P_{A,B}$ is $p_{A,B}$. Since two is invertible, Lemma 2.1 puts the following polynomials in the normalized h -permanental ideal:

$$C_{A,B} = \sum_{\substack{i \in A \\ \{j,\ell\} \subset B}} u_i v_j v_\ell p_{A \setminus \{i\}, B \setminus \{j,\ell\}}, \quad |A| = h - 1, \quad |B| = h, \quad (3.2)$$

together with their transposes. They are homogeneous of Y -degree $h - 2$. Lemma 2.2, applied to these polynomials and the bordered permanents, bounds the dimension of the normalized chart by the dimension of the locus W defined by

$$Y \in V_{h-1}(m, m), \quad C_{A,B} = 0, \quad C_{A,B}^\top = 0. \quad (3.3)$$

Only inclusion in the full ideal of highest parts is needed here; we do not assert that the displayed equations generate that ideal.

First consider the open part of the base where some $r \times r$ permanent $c = p_{R,T}$ is nonzero. When $r = 0$, use the empty permanent $c = 1$, and this is the entire base. Fix Y in this open set, and fix the $2r$ border coordinates u_i for $i \in R$ and v_j for $j \in T$. There remain k coordinates of each border vector. For $i \notin R$ and distinct $j, \ell \notin T$, choose $A = R \cup \{i\}$ and $B = T \cup \{j, \ell\}$ in (3.2). Its highest degree in the remaining border coordinates is exactly

$$c u_i v_j v_\ell. \quad (3.4)$$

Indeed, any other summand uses at least one of the fixed border coordinates and has degree at most two in the remaining ones. The transposed equations similarly have highest parts $c u_i u_{i'} v_j$.

A second application of Lemma 2.2, now in the remaining border variables, and Lemma 2.3 bound every such fiber by k . Letting the fixed $2r$ coordinates vary gives

$$\dim W_Y \leq 2r + k = 2m - k. \quad (3.5)$$

This is a bound for every Y with $c \neq 0$, not just a generic fiber. The finitely many choices of R, T cover the open part under consideration. By induction the base $V_{h-1}(m, m)$ has codimension mk in $\mathbb{A}^{m^2}$. Consequently this part of W has codimension at least

$$mk + k = nk$$

in the normalized ambient space.

If $h \geq 3$, the remaining base is $V_{h-2}(m, m)$. By induction its codimension is $m(k + 1)$. Even allowing all $2m$ border coordinates, the part of W above it has codimension at least $m(k + 1)$. This suffices because

$$m(k + 1) - nk = m - k = h - 2 \geq 0. \quad (3.6)$$

For $h = 2$ there is no remaining base. We have proved that the normalized chart has dimension at most $m^2 + 2m - nk$. Restoring the independent nonzero pivot coordinate adds one dimension, so every nonzero-entry chart of $V_h(n, n)$ has dimension at most $n^2 - nk = n(h - 1)$. These charts cover all points except the origin.

For the reverse inequality, matrices supported on any chosen $h - 1$ rows form an $n(h - 1)$ -dimensional linear space in $V_h(n, n)$. This proves (3.1) and completes the induction. $\square$

Rectangular matrices and arbitrary ground fields. By transposition suppose $a \leq b$. If $h < b$, adjoining $b - a$ zero rows embeds $V_h(a, b)$ as a closed subvariety of $V_h(b, b)$. Every h -permanent involving an adjoined row vanishes, and the others are exactly the original equations. The square case gives $\dim V_h(a, b) \leq b(h - 1)$. Matrices supported on $h - 1$ rows give the opposite inequality.

For a field that is not algebraically closed, Krull dimension of a finitely generated algebra is unchanged by extension to an algebraic closure. The equality and its height formulation therefore hold over every field of characteristic different from two. $\square$

4 Lifting a radical ideal from highest parts

The dimension comparison used above requires no equality of initial ideals. Reducedness does require that equality. We record a sufficient condition, including the case of variables of weight zero.

Lemma 4.1. *Let R be a polynomial ring over a field, equipped with nonnegative integer weights on its variables. For $f_1, \dots, f_s \in R$, let $q_i = \text{in}(f_i)$ be their highest-weight parts. If $q_1, \dots, q_s$ form a regular sequence, then*

$$\text{in}((f_1, \dots, f_s)) = (q_1, \dots, q_s). \quad (4.1)$$

If $(q_1, \dots, q_s)$ is also radical, then so is $(f_1, \dots, f_s)$.

Proof. Set $d_i = \deg f_i$ for the weighted degree. For a representation $F = \sum_i a_i f_i$, let $D = \max_i(\deg a_i + d_i)$. If $D > \deg F$, the weight- D terms give a homogeneous syzygy

$$\sum_i a_i^{[D-d_i]} q_i = 0,$$

where the bracket denotes a homogeneous component. Since the q_i form a regular sequence, their first syzygies are generated by the Koszul syzygies; see [3, Tag 062D]. Thus there are homogeneous $b_{ij} = -b_{ji}$ of weight $D - d_i - d_j$, with $b_{ii} = 0$, such that $a_i^{[D-d_i]} = \sum_j b_{ij} q_j$. Replace a_i by $a_i - \sum_j b_{ij} f_j$. The represented polynomial F is unchanged, because the sum $\sum_{i,j} b_{ij} f_j f_i$ is zero, and the new maximum weight is strictly less than D . Descending through nonnegative integers eventually gives a representation with $D \leq \deg F$. Its highest part lies in (q_i) . This proves the nontrivial inclusion in (4.1); the reverse inclusion is immediate.

Now suppose the latter ideal is radical and $g^N \in (f_i)$ for some $N \geq 1$. The equality of initial ideals gives $\text{in}(g)^N \in (q_i)$, and hence $\text{in}(g) \in (q_i)$. Choose a homogeneous expression for $\text{in}(g)$ in the q_i and replace q_i by f_i . This produces $F \in (f_i)$ with $\text{in}(F) = \text{in}(g)$. The polynomial $g - F$ has smaller weight and represents the same nilpotent residue class. Repetition terminates; at weight zero the subtraction leaves zero. Hence $g \in (f_i)$. $\square$

Lemma 4.2. *Let L be a field, and let $S \in L[z_1, \dots, z_e, x, y, w]$. If its highest total degree part in x, y, w is $cxyw$ for some $c \in L^\times$, then S is squarefree.*

Proof. If a nonconstant irreducible polynomial H occurred at least twice in S , taking highest parts in x, y, w would show that $\text{in}(H)^2$ divides $cxyw$ in the polynomial ring over L . Its only square divisors are units. This also excludes a factor depending only on the z variables, since such a factor would divide the constant coefficient c . Thus H cannot exist. $\square$

5 Reduced complete intersections

Proof of Theorem 1.3. We first work over an algebraically closed field. The case $k = 1$ is the ideal of the two entries of a 1×2 matrix. Suppose $k \geq 2$ and use induction on k . Theorem 1.1 gives height $k + 1$ for $I_k(k, k + 1)$. Since the ideal has $k + 1$ generators in a polynomial ring, these generators form a regular sequence. Its quotient is a complete intersection, Cohen–Macaulay, and pure of dimension

$$k(k + 1) - (k + 1) = k^2 - 1. \quad (5.1)$$

We use the standard regular-sequence and unmixedness properties of Cohen–Macaulay rings; see [3, Tag 00N7].

On a normalized nonzero-entry chart, write

$$X = \begin{pmatrix} Y & u \\ v^\top & 1 \end{pmatrix}, \quad Y \text{ of size } (k - 1) \times k, \quad u \in \mathbb{A}^{k-1}, \quad v \in \mathbb{A}^k.$$

For $j \in \{1, \dots, k\}$ let p_j be the permanent of Y with column j omitted, and let P_j be the maximal permanent of X with that same column omitted. Let R be the maximal permanent omitting the pivot column. Set

$$S = \sum_{i=1}^{k-1} \sum_{1 \leq j < \ell \leq k} u_i v_j v_\ell \text{ per } Y[\{1, \dots, k-1\} \setminus \{i\}, \{1, \dots, k\} \setminus \{j, \ell\}]. \quad (5.2)$$

Lemma 2.1 gives

$$2S = \sum_{j=1}^k v_j P_j - R. \quad (5.3)$$

Thus the normalized chart ideal is $(P_1, \dots, P_k, S)$. For the weights $\deg Y = 1$, $\deg u = \deg v = 0$, these generators have highest parts $p_1, \dots, p_k, S$, respectively.

By induction, the base ring

$$A = K[Y]/(p_1, \dots, p_k)$$

is reduced and a complete intersection, pure of dimension $d = k(k - 2)$. Write $B = \text{Spec } A$. For $k \geq 3$, let $E \subset B$ be the locus where all $(k - 2) \times (k - 2)$ permanents of Y vanish. It is a subvariety of B by expansion of the $(k - 1)$ -permanents. Theorem 1.1 gives

$$\dim E = k(k - 3) = d - k. \quad (5.4)$$

For $k = 2$ take $E = \emptyset$ and use the empty permanent as a nonzero core. In every case E contains no component of B .

At a point of $B \setminus E$, choose a nonzero $(k - 2)$ -permanent c of Y , with row set R_0 and column set T_0 . There is a unique row i outside R_0 and exactly two columns j, ℓ outside T_0 . Regarding the other border coordinates as variables of weight zero, the highest part of S in u_i, v_j, v_ℓ is exactly

$$c u_i v_j v_\ell. \quad (5.5)$$

This follows term by term from (5.2), just as in (3.4).

Consequently S is nonzero over the function field of every irreducible component of B . In the reduced ring $A[u, v]$, an element avoiding all minimal primes is a nonzerodivisor. Hence

$$J = (p_1, \dots, p_k, S) \subset K[Y, u, v]$$

is a complete-intersection ideal, and its quotient is pure of dimension

$$d + (2k - 1) - 1 = k^2 - 2. \quad (5.6)$$

By Lemma 4.2 and (5.5), the polynomial S is squarefree over each of the base function fields. Thus the generic fiber above every component of B is reduced.

We must also exclude components that do not dominate a component of B . Above every point of $B \setminus E$, S is a nonzero homogeneous cubic in $2k - 1$ border variables. Its zero locus has dimension exactly $2k - 2$. Therefore the inverse image of a proper closed subset of B that contains no component, away from E , has dimension at most

$$(d - 1) + (2k - 2) = k^2 - 3.$$

Above E , even allowing the entire border space gives, for $k \geq 3$,

$$\dim E + 2k - 1 = k^2 - k - 1 < k^2 - 2. \quad (5.7)$$

These estimates and purity (5.6) show that every irreducible component of $V(J)$ dominates a component of B . It follows from the reduced generic fibers that $K[Y, u, v]/J$ is generically reduced. A Cohen–Macaulay ring has no embedded associated primes, so generic reducedness implies reducedness; equivalently, use the R_0 and S_1 criterion [3, Tag 031R]. Thus J is radical.

Apply Lemma 4.1 to $(P_1, \dots, P_k, S)$ and $(p_1, \dots, p_k, S)$. Every normalized nonzero-entry chart is reduced. Adjoining the invertible pivot coordinate preserves reducedness, so the original scheme is reduced away from the origin. Its complete-intersection ring has no embedded primes, and every minimal component has positive dimension $k^2 - 1$, so every such component meets a nonzero-entry chart. It is therefore generically reduced, hence reduced everywhere.

Finally, the argument applies over an algebraic closure of every field extension of the original ground field. Faithful flatness of field extensions descends reducedness, while the height and regular-sequence statements have already been proved over arbitrary fields. This proves geometric reducedness and completes the induction. $\square$

6 Consequences and scope

Proof of Corollary 1.2. The partial derivative of per_n with respect to x_{ij} is the permanent of the complementary $(n - 1) \times (n - 1)$ matrix. These complementary matrices comprise all submatrices of that size. The critical locus is therefore $V_{n-1}(n, n)$, whose dimension is $n(n - 2)$ by Theorem 1.1. For each fixed row i , expansion along that row gives

$$\text{per}_n = \sum_{j=1}^n x_{ij} \frac{\partial \text{per}_n}{\partial x_{ij}}.$$

Thus the permanent belongs to its gradient ideal, and the critical locus is also the singular locus of its zero hypersurface. This argument does not divide by n and applies when the characteristic divides the matrix order. $\square$

Corollary 6.1. *With the standard grading, the quotient by $I_k(k, k + 1)$ has Hilbert series*

$$H(t) = \frac{(1 - t^k)^{k+1}}{(1 - t)^{k(k+1)}}.$$

Its multiplicity is k^{k+1} , and its affine scheme is reduced and pure of dimension $k^2 - 1$.

Proof. The $k+1$ generators form a homogeneous regular sequence, each of degree k . The Hilbert-series formula and multiplicity follow by successively quotienting by those nonzerodivisors. Reducedness and purity were proved above. For $k = 1$ the quotient is the ground field and the formula gives $H(t) = 1$. $\square$

For clarity, define the *product count* of a nonzero homogeneous polynomial f of degree at least two as the least s for which

$$f = \sum_{i=1}^s A_i B_i,$$

where A_i, B_i are homogeneous of positive degrees adding to $\deg f$. This is the convention for strength that counts products; the convention with $s + 1$ summands shifts the value by one.

Corollary 6.2. *Over every field of characteristic different from two, the product count of per_n is exactly n for $n \geq 2$.*

Proof. Any such expression implies $V(A_1, B_1, \dots, A_s, B_s) \subseteq \text{Crit}(f)$ by differentiation. The locus on the left is nonempty, since all factors have positive degree, and its codimension is at most $2s$ by the height theorem. For $f = \text{per}_n$, Corollary 1.2 therefore gives $2n \leq 2s$. Expansion along one row expresses per_n as n products of the required kind. The lower bound can be checked after extension to an algebraic closure and so holds over the original field as well. $\square$

Remark 6.3 (Characteristic two). The assumption on the characteristic is essential. In characteristic two, permanents equal determinants. For example, the 2×2 permanents of a 3×3 matrix cut out the matrices of rank at most one, a variety of dimension five rather than three. In (2.3), the coefficient two vanishes, so the additional border equations cannot be deduced. For determinants in other characteristics, the two ordered contributions in that calculation have opposite signs and cancel.

Theorem 1.1 determines dimension, but does not classify components or assert equidimensionality for every $I_h(a, b)$. The reducedness theorem is restricted to $I_k(k, k + 1)$. Neither theorem implies Frobenius purity of all these rings. The results also do not provide a superpolynomial circuit lower bound: the product-count consequence above is linear in n .

A Exact finite checks

The accompanying program `verify.py` verifies identities by expanding permanents as sums over permutations in a sparse polynomial ring over $\mathbb{Z}$. It checks the bordered expansion, (2.3) and its transpose, and the two types of highest parts used in the proofs, for sizes two through five. It also checks every coordinate support in the monomial model (2.4) for $2 \leq k \leq 8$. The latter enumeration comprises 87,376 pairs of supports.

The calculations use exact integer coefficients and no probabilistic substitutions. They check the finite identities and support cases specified by the program. They do not establish dimension, radicality, or the assertions for arbitrary size by finite inference; those claims depend on the proofs in the preceding sections. The program uses only the Python standard library, and the supplement includes its expected output and instructions for reproducing the checks.

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References

- [1] A. Boralevi, E. Carlini, M. Michalek, and E. Ventura, *On the codimension of permanental varieties*, Advances in Mathematics **461** (2025), article 110079. doi:10.1016/j.aim.2024.110079. Preprint version 2: arXiv:2402.17839v2.
- [2] G. A. Kirkup, *Minimal primes over permanental ideals*, preprint (2005), arXiv:math/0510025v1.
- [3] The Stacks Project Authors, *The Stacks Project*, Tag 062D (Koszul regular sequences), Tag 00N7 (Cohen–Macaulay rings), and Tag 031R (reducedness via R_0 and S_1), accessed September 22, 2026.
- [4] Y. Xie, *Frobenius splitting and singular loci of prime-size permanents*, Zenodo preprint (2026). doi:10.5281/zenodo.22878451.
- [5] Y. Xie, *F-purity of the maximal permanental ideal of a generic three-by-four matrix*, Zenodo preprint (2026). doi:10.5281/zenodo.22879129.